

Initial Project Planning Template

Date	14 July 2026
Team ID	
Project Name	HDI Predictor (Human Development Index Predictor)
Team Size	1 (Individual Project)
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Sprint Start Date	Sprint End Date (Planned)
Sprint -1	Environment Setup	USN-1	As a developer, I can set up the Python environment and install NumPy, pandas, Matplotlib, Seaborn, and scikit-learn.	1	High	14 Jul 2026	14 Jul 2026
Sprint -1	Data Understanding	USN-2	As a developer, I can load and inspect the UNDP HDI dataset to understand its columns and missing values.	1	High	14 Jul 2026	14 Jul 2026
Sprint -1	EDA	USN-3	As a developer, I can generate scatter, distribution, correlation heatmap, and strip plots to understand feature relationships.	2	High	14 Jul 2026	14 Jul 2026
Sprint -1	Feature Engineering	USN-4	As a developer, I can clean nulls and log-transform GNI so the model better matches the real HDI formula.	2	High	14 Jul 2026	14 Jul 2026
Sprint -1	Model Building	USN-5	As a developer, I can train and compare Linear Regression and Random Forest, keeping whichever scores higher on R^2 .	3	High	14 Jul 2026	14 Jul 2026
Sprint -1	Model Serialization	USN-6	As a developer, I can persist the trained model with joblib so the app doesn't retrain on every request.	1	High	14 Jul 2026	14 Jul 2026
Sprint -2	Flask Core	USN-7	As a user, I can view a Home dashboard summarizing global HDI stats and EDA charts.	2	High	14 Jul 2026	14 Jul 2026
Sprint -2	Flask Core	USN-8	As a user, I can enter 4 indicators on a Predict page and receive a predicted HDI and tier.	3	High	14 Jul 2026	14 Jul 2026
Sprint -2	Persistence	USN-9	As a user, I can see my past predictions logged on a History page, backed by SQLite.	2	High	14 Jul 2026	14 Jul 2026
Sprint -2	UI/UX	USN-10	As a user, I want form fields to start blank (no auto-filled defaults) so my prediction reflects intentional input.	1	Medium	14 Jul 2026	14 Jul 2026

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Sprint Start Date	Sprint End Date (Planned)
Sprint -3	Deployment	USN-11	As an evaluator, I can access the app on a public URL without running anything locally.	2	High	14 Jul 2026	14 Jul 2026
Sprint -3	Documentation	USN-12	As an evaluator, I can read setup instructions, requirements, and a demo script for the project.	1	Medium	14 Jul 2026	14 Jul 2026

Sprint Summary

Sprint	Focus	Total Story Points	Status
Sprint-1	Data pipeline + model training	10	Completed
Sprint-2	Flask app (3 dashboards) + persistence	8	Completed
Sprint-3	Deployment + documentation	3	Completed